

Q1. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

If $S = 1 - \frac{4}{3} + \frac{9}{3^2} - \frac{16}{3^3} + \frac{25}{3^4} - \frac{36}{3^5} + \dots$, find the value of $160S$.

Q2. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

Find the number of pairs (x, y) satisfying $5x - 19y = 2$ and $x \leq 1000$, where x and y are both positive integers.

Q3. DIRECTIONS *for questions 1 to 3:* Type in your answer in the input box provided below the question.

The average of the marks of five students in a test is 134. The average increases by 23 marks, if the two lowest scoring students are excluded and decreases by 16 marks, if the two highest scoring students are excluded. What is the third highest score among the five students?

Q4. DIRECTIONS for questions 4 to 7: Select the correct alternative from the given choices.

A circle is inscribed in a square, which, in turn, is inscribed in another circle. Find the ratio of the radius of the outer circle to the radius of the inner circle.

☐ a)

$\sqrt{2} : 1$

☐ b) $2 : 1$

☐ c)

$2\sqrt{2} : 1$

☐ d) $4 : 1$

Q5. DIRECTIONS *for questions 4 to 7:* Select the correct alternative from the given choices.

In a 200 m race, A beats B by 4 m and B beats C by 10 m. In a 200 m race, if C and A should finish the race at the same time, what head start should A give C?

- ☐ a) 10 m
- ☐ b) 14 m
- ☐ c) 14.8 m
- ☒ d) 13.8 m ✓ **Your answer is correct**

Q6. DIRECTIONS *for questions 4 to 7:* Select the correct alternative from the given choices.

If the product of all the factors of N is equal to N^8 , which of the following cannot be a factor of N ?

☐ a)

$$2^2 \times 3^1 \times 5^1$$

☐ b)

$$2^3 \times 3^2$$

☐ c)

$$2^6 \times 5^1$$

☐ d)

$$2^2 \times 3^2 \times 5^1$$

Q7. DIRECTIONS *for questions 4 to 7:* Select the correct alternative from the given choices.

A retailer made a profit of 25% by selling an article for ₹120. At what price should he have sold the article, if he wished to make a profit of 50%?

- ☐ a) 124
- ☐ b) 132
- ☒ c) 144 ✓ **Your answer is correct**
- ☐ d) 150

DIRECTIONS *for questions 8 and 9:* These questions are based on the following data.

In a class of 128 students, 100 passed Mathematics, 96 passed Physics, 99 passed Chemistry, 90 passed Biology while 40 passed all the four subjects.

Q8. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

What is the maximum number of students who could have failed in all the four subjects?

DIRECTIONS *for questions 8 and 9:* These questions are based on the following data.

In a class of 128 students, 100 passed Mathematics, 96 passed Physics, 99 passed Chemistry, 90 passed Biology while 40 passed all the four subjects.

Q9. DIRECTIONS *for questions 8 and 9:* Type in your answer in the input box provided below the question.

What is the minimum number of students who could have passed exactly two subjects?

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

In a sequence of 400 numbers, the product of any two consecutive terms is 20, and the 59th term is 15. If the sum of the terms from the 278th term to the 378th term is equal to S , find the value of $300S$.

Q11. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

The sides AB, BC and CA of a triangle ABC are 8, 6 and 4 respectively and the incircle of triangle ABC touches the sides AB, BC and CA at P, Q and R respectively. What is the ratio of the lengths of AP and CR?

☐ a) 1 : 3

☐ b) 3 : 1

☐ c) **2 : 3**

☐ d) **3 : 2**

Q12. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

Find the ratio of x and y , if $2x - 7y + t = 0$ and $5x - 3y - 2t = 0$.

☐ a)

$$\frac{3}{5}$$

☐ b)

$$\frac{17}{9}$$

☐ c)

$$\frac{10}{7}$$

☐ d) **Cannot be determined**

Q13. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

Consider the following two curves in the xy plane: $y = x^4 + x^2 + 5$ and $y = x^2 + x + 6$. If the two curves intersect at n points, which of the following is true?

- ☐ a) $n = 0$
- ☐ b) $n = 1$
- ☐ c) **$n = 2$**
- ☐ d) **$n = 3$**

Q14. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

If an arithmetic progression has the sum of its first n terms as zero but none of the terms is zero, which of the following statements is true?

- ☐ a) n is always even.
- ☐ b) n is always odd.
- ☐ c) n is always a multiple of 4 or 5.
- ☐ d) More than one of the above

Q15. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

Two persons, A and B, are approaching the foot of a tower, from opposite directions, at same speeds. If at a certain point of time, the angles of elevation of the top of the tower as seen by A and B are 60° and 30° respectively, and A took 15 minutes from that moment to reach the foot of the tower, then after how much more time (in minutes) will B do the same?

☐ a) 15

☐ b) 30

☐ c) **45**

☐ d) 60

Q16. DIRECTIONS for questions 11 to 18: Select the correct alternative from the given choices.

If P, Q and R are distinct positive real numbers, $\frac{P^4 (Q^2 + R^2) + Q^4 (P^2 + R^2) + R^4 (P^2 + Q^2)}{(PQR)^2}$ must be more than

- ☐ a) 8.
- ☐ b) 7.
- ☐ c) **6.**
- ☐ d) 5.

Q17. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

Two painters, Arjo and Bulu, together got a contract to paint a house for ₹6000. On the day of the work, Arjo started the work as scheduled but Bulu turned up one hour late and as a result they took 45 minutes more to complete the job. What would be Bulu's share in the total amount had they both worked for an equal amount of time?

☐ a) ₹2400

☐ b) ₹3600

☐ c) ₹4000

☐ d) ₹4500

Q18. DIRECTIONS *for questions 11 to 18:* Select the correct alternative from the given choices.

If x , y and z are odd integers, which of the following statements is not true?

- ☐ a) $(x^2y + y^2z + z^2x)xyz$ is odd
- ☐ b) $(xy + yz + zx)(x + 2y + 3z)$ is even
- ☐ c) $(xy + yz + zx)(2x + y)^2$ is odd
- ☐ d) $(2x + 3y + 4z)(xyz + 6)$ is even

Q19. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

If $\frac{(x)_8}{(10)_{10}} = (40)_8$, what is the value of x ?

Q20. DIRECTIONS *for questions 19 and 20:* Type in your answer in the input box provided below the question.

In a chess tournament, every person played one game with every other person participating in the tournament. The total number of games that men played between themselves exceeded those played by men with women by 18. If there were 4 women in the tournament, how many games in total were played in the tournament?

Q21. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.

Find the distance between the lines $5x + 12y + 21 = 0$ and $15x + 36y + 297 = 0$.

☐ a) 3

☐ b) 6

☐ c) 13

☐ d) 39

Q22. DIRECTIONS *for questions 21 and 22:* Select the correct alternative from the given choices.
How many four-digit numbers divisible by 9 can be formed by using exactly two distinct digits?

- ☐ a) 26
- ☐ b) 42
- ☐ c) 69
- ☐ d) 56